

機械学習を用いた因果推論

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因果推論 at glance

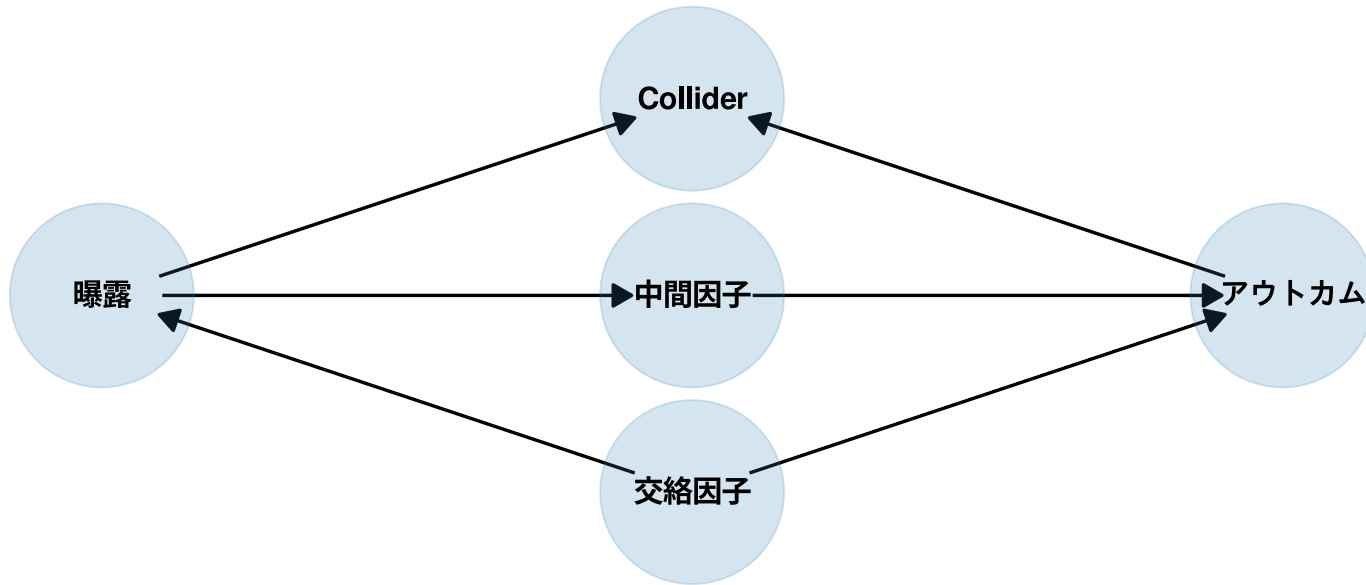
因果関係

- ▶ 医学研究の第一の目標と言っても過言ではない
- ▶ A の薬を使うと患者の予後は良くなるか？
- ▶ 実際は口で言うほど簡単ではない

因果関係の考え方

- ▶ 大きく分けて 2 つの考え方がある
 - 潜在的アウトカムを考える Rubin 流
 - DAG を作り、do operator を使う Pearl 流
- ▶ 詳細は省くが、今回は Pearl 流の DAG を中心に考える

DAG の基本



通常 of 統計学的手法

- ▶ 通常は交絡因子の調整により因果関係を推測する

交絡因子の調整の方法

- ▶ 回帰
- ▶ Matching
- ▶ 層別化

例えば単純な回帰だと

term	Coef	Std.Err	t	P
Intercept	-1174.13	2456.31	-0.48	0.63
treat	1548.24	781.28	1.98	0.05
age	12.98	32.49	0.4	0.69
educ	403.94	158.91	2.54	0.01
race_hispan	1739.54	1018.52	1.71	0.09
race_white	1240.64	768.76	1.61	0.11
married	406.62	695.47	0.58	0.56
nodegree	259.82	847.44	0.31	0.76
re74	0.3	0.06	5.09	0
re75	0.23	0.1	2.21	0.03

Matching だと

- ▶ 一番有名なのは Propensity score matching

```
treat age educ ... race_hispan race_white propensity_score
19    1  26  12 ...         0         0         0.619048
42    1  17   8 ...         0         0         0.619048
156   1  18   9 ...         0         0         0.217391
111   1  17   9 ...         1         0         0.217391
148   1  26  11 ...         0         0         0.217391
```

[5 rows x 11 columns]

matched size: (296, 11)

ATT (diff in means on matched sample): 1008.6141578378374

Doubly robust estimation

DR ATE: 1394.87

SE: 926.77

95% CI: (-421.60, 3211.33)

ここらへんの問題

- ▶ Misspecification の問題
- ▶ Estimand の問題 = 治療効果の異質性

機械学習 at glance

代表的な機械学習

- ▶ Logistic 回帰
- ▶ tree-based model
 - Random forest
 - XGBoost
- ▶ Neural network

疑問

- ▶ 機械学習を用いて因果関係を推測出来るんじゃないか？

機械学習を用いた因果推論の Pros and Cons

▶ Pros

- モデルの Misspecification を避けられる
- 因果関係の異質性を柔軟に捉えられる

▶ Cons

- Overfitting の問題
- 解釈が難しい時がある
- 信頼区間が出しにくい

```
<econml.metalearners._metalearners.SLearner object at 0x7df465aa73b0>
```

```
[-1530.55967406  369.38136095 -1198.06865549  -86.11625054  
-833.38280696  171.70918773   0.          171.70918773  
 319.60055729  3232.14589762 -1326.64678003   62.06342827  
 973.47414377  503.67373985 -498.5500903  -128.93867654]
```

機械学習を用いた因果推論の Pros and Cons

```
485.25799913 489.9931725 1943.89516126 -269.05807531
2651.3162438 251.02443635 -3528.65968649 -591.89366066
436.53410022 32.09860508 -149.43314728 214.47455118
-86.11625054 32.09860508 3232.14589762 32.09860508
664.43858698 215.4353309 3451.76242614 -3558.48624607
-1148.88963224 2872.53054803 524.46483345 -544.94289508
632.3399819 -545.38598944 2651.3162438 144.60637046
-20.61071748 95.79529433 -86.11625054 -544.94289508
179.0257057 -2760.46335128 -957.9515668 1897.03733375
650.11851089 189.4222147 29.11044322 848.41449744
-352.45433824 2642.76190818 -1788.65634271 -988.00134014
-1198.06865549 3232.14589762 -522.42034513 -325.05097239
32.09860508 -167.70521158 2297.82108649 18.95269302
-1323.72233293 -35.38360418 74.66266901 -263.26905945
-289.37725534 3185.71468272 -122.99846869 -86.11625054
-128.93867654 1179.17631607 -2787.07045179 1789.97076026
32.09860508 -263.26905945 -843.00231034 -1539.13799559
```


機械学習を用いた因果推論の Pros and Cons

```
125.67766335 425.69190007 2199.59715977 -2259.89874238
2651.3162438 -1539.13799559 261.5062125 -495.49910709
-1218.45302373 -1539.13799559 2297.82108649 278.45737341
1861.49985719 664.43858698 95.79529433 2420.81146399
903.14992197 -329.34261136 894.79147023 194.23270811
-1179.12813405 171.70918773 171.70918773 -1089.00156337
341.10564214 369.38136095 -591.89366066 -86.11625054
-1131.29106563 -410.70666473 -20.61071748 -591.89366066
207.64243233 -86.11625054 319.60055729 2063.90546808
-2635.07325719 -662.79260714 715.06165279]
```

Scratch for S-learner

モデルの学習を開始します...

```
LGBMRegressor(random_state=42, verbosity=-1)
```

学習が完了しました。

テストデータのRMSE: 0.6124

平均処置効果の推定値: 749.0242

	treat	age	married	educ	...	effect_0	treat_1	effect_1	treat_e	>
0	1	37	1	11	...	2437.328621	1	6254.264565	3816.9	>
1	1	22	0	9	...	5855.361897	1	6215.004201	359.6	>
2	1	30	0	12	...	1654.736704	1	5863.334241	4208.5	>

Scratch for S-learner

```
3  1 27  0 11 ... 5778.341560  1 10567.286106 4788.9 >
4  1 33  0  8 ... 895.305362   1 3814.387914 2919.0 >
```

>

[5 rows x 15 columns] >

```
> ffct
```

```
> 35943
```

```
> 42304
```

```
> 97538
```

```
> 44546
```

```
> 82552
```

```
>
```

```
>
```

T learner

```
<econml.metalearners._metalearners.TLearner object at 0x7df466952990>
```

T learnerでの局所治療効果

```
[ -6650.98045077 -1118.6693171  7910.85611871 -4644.60006793  
-5193.61671006 -1265.01628653 -1076.66778986 -1884.26404861  
1564.51358604  6387.09977938 -17618.45105122 -3612.68936737  
 573.13368963 16200.95698813 -3859.86899084  336.54393078  
2849.52966914  5133.3472875  4362.02429344 -4925.08721332  
-3299.94537519 -13131.62022045 -12833.90437168 -8136.22119316  
2871.03924138 -1936.90571189 -10729.46961207 -550.75806049  
-4162.5337728 -16485.71427559  4827.58362554 -9824.65040004  
8204.3069157  1938.90093947 11337.47819452 -3406.60012456  
-3963.31021415 2860.03677434  6471.99906902 -6261.968898  
917.83931442 -1937.86327265 13587.22837397 1821.31726953  
865.44040096 2312.87813147  934.3501449 -16426.57428451
```

T learner

594.8779234	3585.2194147	-4277.95596726	2136.27681753
7125.14806731	2138.29716778	-3883.91378688	-5218.15875866
-7799.68434228	4075.03162163	-5429.40421769	-9084.61075982
-8857.81565801	5029.10118391	-9352.85425194	-267.04791813
9704.22887422	-4425.27599461	2037.97086778	-9330.43734666
3460.91088841	5143.14296934	706.29492043	-8247.97419454
-621.94188682	9497.28415412	7091.82273943	-4162.5337728
-365.05015182	4028.98226432	4498.00494588	4616.20332428
-4302.66307485	-831.07548236	-3761.67302329	-4863.79074393
2338.24688147	8510.37044263	-4976.29005251	-3629.85675494
7007.06944427	-5855.50651842	4731.59503049	-3063.50546107
-912.18517109	-15568.23504452	2641.3341303	-5453.70340661
10867.85011588	5753.16004211	-1456.64247255	3368.49826808
-3180.22245828	2500.74116673	4251.33460448	-1130.11883784
-803.66883908	-3089.84565141	-2072.82922894	1787.99070559
-1960.36916504	4389.53911275	-9042.29350198	-4162.5337728
117.86381757	730.53927385	-3415.59825	-4492.28134015

T learner

```
-5048.95189277 -4162.5337728 3750.17902964 2291.90717807  
3355.0384531 589.82480352 2833.65279389]
```

```
<econml.metalearners._metalearners.XLearner object at 0x7df464dd4650>
```

```
<econml.metalearners._metalearners.DomainAdaptationLearner object at 0x7df465a >
```

```
> a7b00>
```

```
<econml.dr._drlearner.DRLearner object at 0x7df4624b9bb0>
```

機械学習の因果推論における使い方

- ▶ 交絡因子の調整で機械学習を用いる
 - AIPW, tlme など
- ▶ 因果をダイレクトに推論する学習機を作る
 - Meta-learner
- ▶ CATE を測定する
 - Causal forest など

因果 forest

- ▶ HTE を計測するのに有名
- ▶ Ashley et al 2019

Survival model

	id	study	rx	sex	age	obstruct	perfor	adhere	nodes	status	differ	extent
1	1	1	Lev+5FU	1	43	0	0	0	5	1	2	3
2	1	1	Lev+5FU	1	43	0	0	0	5	1	2	3
3	2	1	Lev+5FU	1	63	0	0	0	1	0	2	3
4	2	1	Lev+5FU	1	63	0	0	0	1	0	2	3
5	3	1	Obs	0	71	0	0	1	7	1	2	2
6	3	1	Obs	0	71	0	0	1	7	1	2	2

	surg	node4	time	etype
1	0	1	1521	2
2	0	1	968	1
3	0	0	3087	2
4	0	0	3087	1
5	0	1	963	2
6	0	1	542	1

Survival model

```
      control treated  
[1,] -625.9972  625.9972  
[2,]  140.6130 -140.6130  
[3,] -584.5790  584.5790  
[4,] -576.2461  576.2461  
[5,] -184.8944  184.8944  
[6,] -1499.8869 1499.8869
```

Survival model

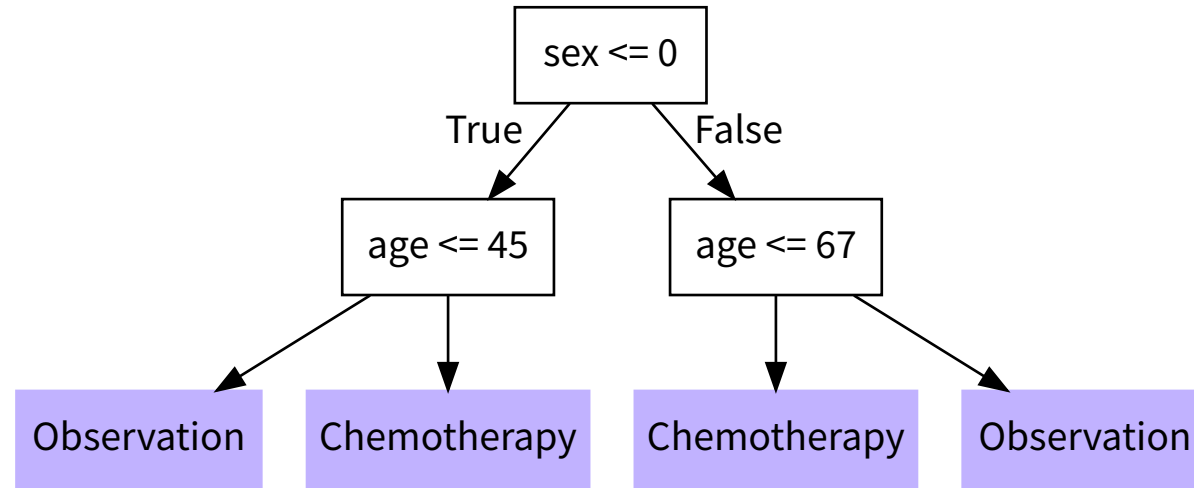


Figure 1: Treeplot